No confirmed extragalactic planets:
a reassessment of every published candidate

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Abstract

Eleven claims of planets, or planet-mass populations, beyond the Milky Way have appeared since 1996, together with three claims of planets around Milky Way stars of extragalactic origin. None has been confirmed, and none can be repeated: each rests on a single unrepeatable event or on a statistical inference whose null hypothesis was never computed. We reassess all of them with a common method: reconstruct the published measurement, replace the authors' favourable assumption with the full prior, condition the inference on the fact that the event was found, and compute the null model. For M51-ULS-1b the X-ray dip does not measure the occulter's radius (a grazing star fits within $\Delta \ln L = 0.7$ of a central Saturn), the achromaticity test carries a Bayes factor of only $\sim$ 2–8, and once detection is conditioned on, the most probable occulter is an M-dwarf tertiary; the event rate also exceeds that expected from bound companions by 10^2–10^3. For PA-99-N2 the "$6\,M_{\rm J}$ planet" requires a $0.5\,M_\odot$ lens in the M31 disc, which implies a source size $\rho \simeq 0.10$–0.20 Einstein radii, whereas the sharpness of the fitted anomaly requires $\rho < 0.021$; forcing $\rho = 0.10$ in a finite-source binary-lens refit degrades χ^2 by 65–154 (2% errors), and a rate-weighted lens-population analysis gives a 2–5% probability of a planet in M31. The Fe Kα line-shift statistics behind the "$\gtrsim$ 2000 planets per star" claims in three lensed quasars are reproduced by a static spectrum containing the broad relativistic line that RX J1131−1231 is known to possess, fitted with narrow Gaussians in low-count spectra. The FRB 20180916B planet fails a beaming-independent energy budget by $10^{3.7}$–10^7; the published BD+20 2457 two-companion system disintegrates in 4–2500 yr; the NGC 4845 tidal-disruption "super-Jupiter" is one corner of a victim-mass degeneracy spanning $2\,M_{\rm J}$ to $3\,M_\odot$; the Twin-Quasar and Einstein-Cross "planets" follow from misreadings of event timescales. We identify four recurring failure modes and state the measurements a future confirmation would need.

1 Introduction

Planets are common in the Milky Way, and nothing distinguishes the stars of other galaxies in a way that should prevent them from hosting planets. Finding one, however, is a different matter. At megaparsec distances individual stars cannot be resolved, radial velocities and transits are out of reach, and every proposed detection method relies on a single, unrepeatable event: a microlensing anomaly, an X-ray eclipse, a brightness fluctuation in a lensed quasar, a flare from a disrupted body.

The literature now contains eleven such claims beyond the Milky Way, catalogued in the usual reference lists (e.g. Wikipedia, 2026): M51-ULS-1b, an X-ray transit candidate in the Whirlpool galaxy (Di Stefano et al., 2021); PA-99-N2, a binary-lens microlensing event towards M31 interpreted as a star–planet pair (An et al., 2004; Ingrosso et al., 2009, 2011); free-floating planet populations in the lens galaxies of RX J1131−1231, Q J0158−4325 and SDSS J1004+4112 inferred from Fe Kα line shifts (Dai & Guerras, 2018; Bhatiani et al., 2019); a "3 Earth-mass" body in the Twin Quasar lens (Schild, 1996; Colley & Schild, 2003); a 1.6 Jupiter-mass reading of the 1989 Einstein-Cross event (Irwin et al., 1989) and a 2024 "nanolens" population in the same system (Tuntsov, 2024); a $2.9\,M_\oplus$ planet orbiting the magnetar responsible for FRB 20180916B (Kurban et al., 2022); and a 14–40 $M_{\rm J}$ object destroyed in the tidal disruption event in NGC 4845 (Nikołajuk & Walter, 2013; Lei et al., 2016). Three further claims concern planets around Milky Way stars thought to have formed in accreted dwarf galaxies: HIP 13044 b (Setiawan et al., 2010), BD+20 2457 b and c (Niedzielski et al., 2009; Perottoni et al., 2021) and a planet-engulfment inference for HD 134440 (Reggiani & Meléndez, 2018).

None of these is listed as confirmed in the NASA Exoplanet Archive, and the discovery papers themselves use the word "candidate". Yet they are routinely described in secondary literature as "the first planets in other galaxies". In this paper we ask a narrower question of each: given the data that exist, does the planetary interpretation survive a complete analysis? We find that it does not, in any case. Several claims are internally inconsistent with the data that produced them; others rest on a null hypothesis that was never computed; others follow from a favourable choice among unconstrained nuisance parameters; and two had already been withdrawn.

Section 2 describes the common method. Sections 3–9 treat the candidates in turn, in decreasing order of the effort each required. Section 10 draws out the common failure modes and specifies what a confirmation would need. Section 11 tabulates the outcome.

2 Method

2.1 Single-event inference

Every candidate is a single event. For such data the posterior on the physical parameter of interest, θ (a radius, a mass, a location), is

$$p(\theta \mid d,\ \text{found}) \propto \mathcal{L}(d \mid \theta)\, P_{\rm det}(\theta)\, \pi(\theta), \qquad (1)$$

where $\mathcal{L}$ is the likelihood of the light curve or spectrum, π is the astrophysical prior over the populations that could have produced the event, and $P_{\rm det}$ is the probability that an event with parameters θ would have been found by the search that found it. The discovery papers examined here evaluate $\mathcal{L}$, sometimes with care, but replace $P_{\rm det}\,\pi(\theta)$ by a choice: a uniform prior over a convenient range, or a single assumed value for the parameter that sets the scale (the lens mass, the black-hole mass, the number of debris clumps). When $\mathcal{L}$ is nearly flat in θ, which is the generic situation for a single event, that choice *is* the result. Our first step in each case is therefore to map $\mathcal{L}$ over the full range of θ and to restore $P_{\rm det}$ and π.

2.2 Null models

Several claims are statistical: an excess frequency of some feature is attributed to planets because a model without planets predicts fewer. The decisive question is then what the model *without the claimed effect* predicts once all known ingredients are included, calibrated with the look-elsewhere effect appropriate to the search. We compute these nulls by simulation.

2.3 Reconstruction from published parameters

The raw data of several candidates (Chandra event files, POINT-AGAPE photometry) could not be retrieved from the environment in which this analysis was carried out. Where that is so we reconstruct the measurement from the published model definition and fitted parameters, using "Asimov" data sets (expected values, no noise) and verifying that our likelihood reproduces the published posterior before any new analysis is layered on top. Conclusions drawn this way concern the information content of the published fit, which is what the published claims rest on.

2.4 Independent verification

The load-bearing computations—the M51-ULS-1b radius degeneracy, the PA-99-N2 finite-source refit, the Fe Kα line-shift simulation, the FRB energy budget and the BD+20 2457 integrations—were re-derived from scratch by an independent implementation that had access to the specification of each calculation but not to our code. Where the first pass of our own analysis contained an error (an unequal treatment of blending in the PA-99-N2 refit, §4.2), the verification caught it and the corrected numbers are the ones reported.

3 M51-ULS-1b

3.1 The claim

Di Stefano et al. (2021) searched 2624 Chandra light curves of 238 bright X-ray sources in M51, M101 and M104 for intervals in which the count rate fell to zero and recovered, and found one: a $\simeq 3$ h symmetric dip in the ultraluminous supersoft source M51-ULS-1 during ObsID 13814 (2012 September 20), with 20–30 min at zero flux. Fitting an opaque disc crossing a uniform X-ray photosphere of radius $R_{\rm X} = 2.5^{+4.1}_{-1.1} \times 10^9$ cm (Urquhart & Soria, 2016), they obtained an occulter radius with mode $0.74\,R_{\rm J}$ (90% interval 0.18–$2.7\,R_{\rm J}$), a transverse speed of $17\,{\rm km\,s^{-1}}$ (5–$56\,{\rm km\,s^{-1}}$) and hence a circumbinary orbit of tens of AU, and argued from the constancy of the X-ray hardness ratio across the dip that it is an eclipse rather than absorption or a state change. Stellar companions were excluded on the grounds that the system is younger than 20 Myr and a star of that age is larger than the fitted radius.

3.2 The light curve does not measure the radius

We reimplemented the published model (five parameters: baseline counts $c_{\rm X}$, mid-time, impact parameter b, radius ratio $f = R/R_{\rm X}$, speed v; 471.156 s bins; Poisson likelihood; background 0.38 counts per bin) and fitted an Asimov light curve constructed from the published values ($c_{\rm X} = 7.6$, 10.5 ks first-to-last contact, 1.56 ks at zero). Restricted to $f < 3$ the posterior reproduces Table 1 of Di Stefano et al. (2021): radius mode $0.73\,R_{\rm J}$, speed mode $18\,{\rm km\,s^{-1}}$.

Profiling the likelihood over f with the other parameters free reveals why that restriction matters (Fig. 1). An opaque body of *any* radius $R \geq R_{\rm X}$ whose limb passes through the source, $b \simeq R - R_{\rm X}$, fits the dip to within $\Delta \ln L = -0.13$ ($f = 2$), -0.48 ($f = 5$), -0.61 ($f = 10$) and -0.70 ($f = 30$, i.e. $R \simeq 1.3\,R_\odot$). The grazing geometry trades a faster limb crossing for a larger body and reproduces the ingress, the flat bottom and the egress. A likelihood ratio of 2 is no discrimination at all. The published "Saturn-sized" mode arises because the prior on b is uniform out to $51R_{\rm X}$ while the ridge of acceptable large-f solutions is only $\sim 0.1R_{\rm X}$ wide in b; with a prior proportional to the *transiting* range of b, which is the physically correct one, the ridge is not penalised and the 90% interval on R becomes 0.14–$14\,R_{\rm J}$. A generic trapezoid or a flat-bottomed Gaussian fits as well as the occultation geometry ($\Delta \ln L = 0.04$), so the dip shape carries no evidence for a transit as opposed to any other symmetric interruption.

3.3 The achromaticity test is weak

The argument that a constant hardness ratio excludes absorption and photospheric changes was qualitative. We folded a blackbody of $kT = 70$–130 eV through

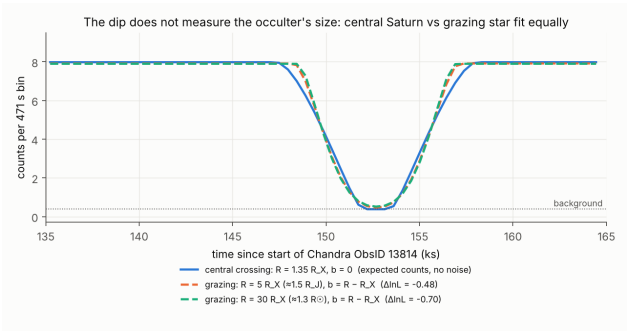


Figure 1: Expected counts for the published M51-ULS-1b solution (a body of $1.35R_{\rm X}$ crossing the source centrally) and for grazing occulters of $5R_{\rm X} \simeq 1.5\,R_{\rm J}$ and $30R_{\rm X} \simeq 1.3\,R_\odot$, with the other parameters re-optimised. The three curves differ by $\Delta \ln L \leq 0.7$ for the observed data quality.

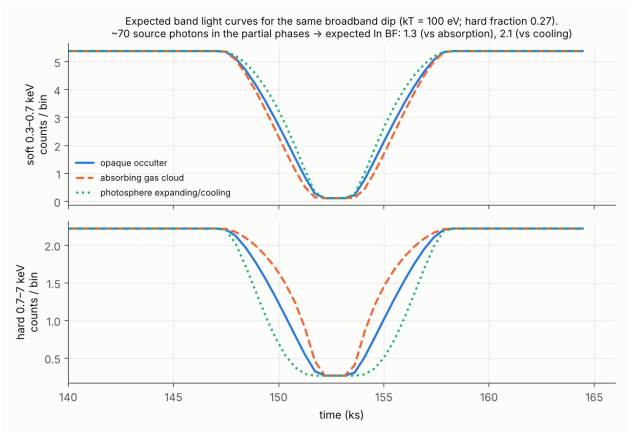


Figure 2: Soft and hard light curves expected for the same broadband dip under the three hypotheses, $kT = 100$ eV. The curves differ by a few counts.

an approximate 2012 ACIS-S response (with Morrison–McCammon absorption) and constructed, for the same broadband dip, the soft (0.3–0.7 keV) and hard (0.7–7 keV) light curves expected from (A) an opaque occulter, (B) an absorbing cloud whose column reproduces the broadband depth at each instant, and (C) an expanding, cooling photosphere at constant bolometric luminosity, the mechanism of the X-ray off-states this class of source is known to display (Greiner & Di Stefano, 2002). With ~ 70 source photons in the partial phases and a hard fraction of 0.27 at $kT = 100$ eV, the expected log Bayes factor for A over B is 1.3 and for A over C is 2.1 (Fig. 2); across the temperature range it spans 0.3–4.2. The constancy of the hardness ratio favours an opaque body by odds of roughly 2:1 to 8:1.

3.4 Detection-conditioned inference

A transit found in a search must be weighted by the probability of being found: the alignment probability $\propto (R_{\rm X} + R)/(\delta\, a)$, where $\delta \simeq 0.15$ rad is the inclination spread allowed by the donor eclipses the system shows, and the timing probability $T_{\rm obs}/P_{\rm orb} \propto a^{-3/2}$. We re-weighted the light-curve posterior by these factors and by physical priors over four classes of occulter at the system's age of 5–20 Myr: sub-Saturns (0.35–1.2 $R_{\rm J}$), giant planets (1.1–1.8 $R_{\rm J}$), brown dwarfs (1.5–5.4 $R_{\rm J}$) and M dwarfs (3.5–15 $R_{\rm J}$), with orbits between the circumbinary stability limit $2.4\,a_{\rm bin}$ (Holman & Wiegert, 1999) and 300–1000 AU, $M_{\rm tot}$ between 10 and 40 $M_\odot$ and eccentricities up to 0.5. The detection-conditioned posterior moves every class inward and the mixture towards larger bodies: a giant planet at 6^{+6}_{-2} AU ($P \simeq 4$ yr), a brown dwarf at 2.6 AU, an M dwarf at 1.7 AU (Fig. 3). With occurrence priors of 0.2, 0.1, 0.05 and 0.15 per massive binary, the class probabilities *given* an occultation are 1%, 7%, 13% and 79%; tripling the planet priors raises the planetary share to 21%.

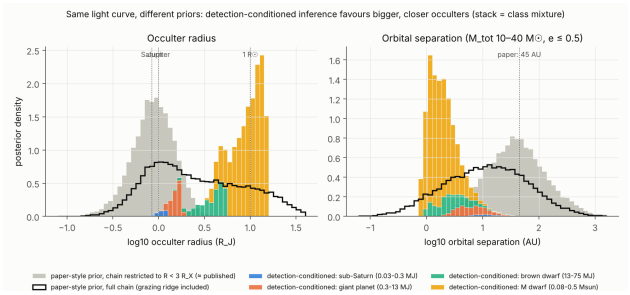


Figure 3: Occulter radius and orbital separation for M51-ULS-1b under the published prior (grey; black outline when the grazing ridge is sampled) and under the detection-conditioned class mixture (stacked).

3.5 Occultation or intrinsic variability: the base rate

Summed over the four classes, bound-body occultations of a bright X-ray binary are expected at 2.4×10^{-5} per source-megasecond. The search found one in 20–200 source-Ms (the exposure actually sensitive to a 1 ks drop to zero), a factor 10^2–10^3 above expectation. Di Stefano et al. (2021) resolve this by inferring "$64/g$" wide-orbit companions per X-ray binary with $g \simeq 1$, i.e. every companion's orbit aligned with the line of sight to $\sim 10^{-5}$ rad; the donor eclipses constrain the inclination only to ~ 0.1 rad, and circumbinary mutual inclinations are $\gtrsim 0.01$ rad. The alternative is that the dip is not an occultation. This source shows off-states, energy-dependent double dips and stellar eclipses in the same archive; if intrinsic or absorption events that mimic a 3 h achromatic dip occur at even 10^{-3} per source-Ms, the probability that this dip is an occultation is $\simeq 10\%$ at the central Bayes factor (Fig. 4). We also checked all on-state windows of the 2000–2018 archive for a repeat at $T_0 + nP$: only 2% of the short-period posterior is excluded, so the archive is uninformative.

3.6 Verdict

M51-ULS-1b is a 3 h achromatic dip in a variable supersoft source, consistent with an opaque body of any

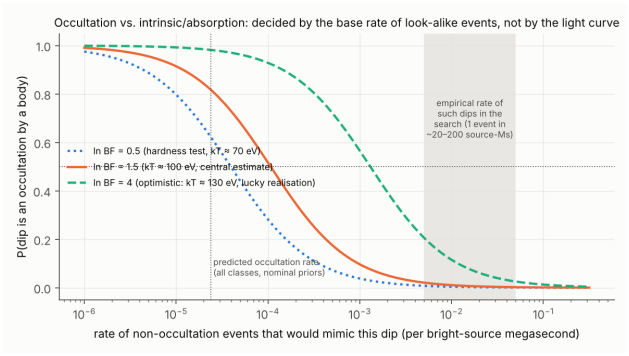


Figure 4: Probability that the M51-ULS-1 dip is an occultation by a body, as a function of the rate of look-alike intrinsic or absorption events, for three values of the shape/hardness Bayes factor. The vertical line is the predicted bound-body occultation rate; the shaded band is the empirical rate of such dips in the search.

radius $\gtrsim 0.3\,R_{\rm J}$. Given an occultation, a planet is the least probable class; whether it is an occultation at all depends on a base rate the data do not constrain. More Chandra time on this source cannot settle it: with $P \simeq 0.5$–10 yr the chance of a repeat per Ms is 0.003–0.1, and a repeat at this count rate leaves the size degeneracy intact. Separating a central $1.35R_{\rm X}$ body from a grazing star at $\Delta \ln L \simeq 5$ requires about seven times the photons in the partial phases.

4 PA-99-N2

4.1 The claim

PA-99-N2 is a bright 1999 pixel-lensing event 22′ from the centre of M31 (Paulin-Henriksson et al., 2003). An et al. (2004) showed that its light curve departs from a point-lens model near peak and is fitted by a binary lens with $t_{\rm E} = 125$–132 d, mass ratio $q = 10^{-1.85}$ (close solution C1), $10^{-1.91}$ (wide, W1) or $10^{-2.12}$ (finite-source solution FS), and found marginal finite-source effects, $\rho = (1.31 \pm 0.30) \times 10^{-2}$ with $\Delta\chi^2 = 4.9$ relative to a point source and $\rho < 2.06 \times 10^{-2}$. The source is an M giant in the M31 disc with angular radius $\theta_* = 0.51 \pm 0.06\,\mu$as. An et al. (2004) concluded that the lens is most probably in the M31 halo and "most probably a stellar remnant". Ingrosso et al. (2009, 2011) instead adopted a $0.5\,M_\odot$ lens in the M31 disc, the most probable mass for disc self-lensing, and announced a $6.34\,M_{\rm J}$ companion.

4.2 The finite-source constraint excludes the assumed lens

Microlensing fixes $\theta_{\rm E}^2 = \kappa M \pi_{\rm rel}$ with $\kappa = 8.144$ mas $M_\odot^{-1}$. For a lens inside M31 the relative parallax is tiny, so a $0.5\,M_\odot$ star 1–4 kpc in front of the source has $\theta_{\rm E} = 2.6$–$5.2\,\mu$as and the source subtends $\rho = \theta_*/\theta_{\rm E} = 0.10$–0.20 Einstein radii, five to ten times the published upper limit (Fig. 5, left). Equivalently, the limit $\theta_{\rm E} \gtrsim 25\,\mu$as requires $M \gtrsim 114\,M_\odot\,(1\,{\rm kpc}/D_{\rm LS})$ for an M31 lens and, with $t_{\rm E} = 127$ d, a lens–source relative speed $\gtrsim 260\,{\rm km\,s^{-1}}$ (Fig. 5, right); disc stars move at $\sim 60\,{\rm km\,s^{-1}}$ relative to one another.

Because the finite-source fit of An et al. (2004) was a restricted search around C1 and was flagged as possibly affected by systematics, we tested the constraint directly. We computed finite-source binary-lens magnification maps for the C1 geometry by inverse ray shooting (81 million rays, $0.0015\theta_{\rm E}$ pixels; companion on the side that reproduces the published anomaly, $\sim 14\%$ peak-to-peak 3.6 d before t_0), generated the $\rho = 0.013$ light curve with POINT-AGAPE-like sampling (167 epochs, a gap at JD′ 72–91) and 2% errors, and refitted it with models forced to larger ρ, letting t_0, u_0, $t_{\rm E}$, α and the fluxes float in a global search, with the baseline pinned to within $-10\%/+50\%$ of the source flux as the flat 2000–2001 seasons pin it. Across four epoch realisations the best attainable χ^2 is 52–93 at $\rho = 0.05$, 65–154 at $\rho = 0.10$ and 109–201 at $\rho = 0.20$, against 0 for the published solution and 134–161 for a single lens (Fig. 6). A disc-star source size does not smooth the fitted anomaly; it produces a different, 25-day-wide feature. Scaled to the real photometric precision these penalties are several hundred in χ^2. An independent implementation confirmed the light curves and the refits; it also exposed an error in our first pass, which had compared a flux-constrained binary against a flux-free single lens, and showed that with unconstrained negative blending both models find unphysical shortcuts. The numbers above are the corrected ones.

4.3 Rate-weighted lens-population inference

We built a Bayesian model with five lens populations—M31 disc, spheroid and halo, Milky Way halo and Milky Way disc—normalised by the optical depths of An et al. (2004) (halo MACHO fractions 1–20%) and a local-density model for the Galactic disc; a present-day mass function including white dwarfs, neutron stars and black holes (Kroupa IMF, initial–final mass relations); line-of-sight density profiles; realistic kinematics (disc dispersion plus shear, spheroid and halo dispersions, the source's transverse rotation, solar motion); a luminosity veto on foreground main-sequence lenses brighter than $V = 22$; and the measured $t_{\rm E}$ and θ_*. Each population was sampled τ-weighted and converted to event-rate weights, $\Gamma = (2/\pi)\tau\langle 1/t_{\rm E}\rangle$. The finite-source information was applied three ways: ignored, as an upper limit only, and as published (Table 1, Fig. 7).

Ignoring the finite-source information reproduces the published picture: a $0.6\,M_\odot$ M31 disc star with a $\sim 9\,M_{\rm J}$ companion. That picture predicts $\rho \simeq 0.1$, which the light curve excludes. With the published constraint, the lens is in M31 with probability 0.8–0.95 but is a 5–50 $M_\odot$ remnant in the halo or spheroid moving at $\sim 400\,{\rm km\,s^{-1}}$, with a $\sim 0.1\,M_\odot$ companion; the

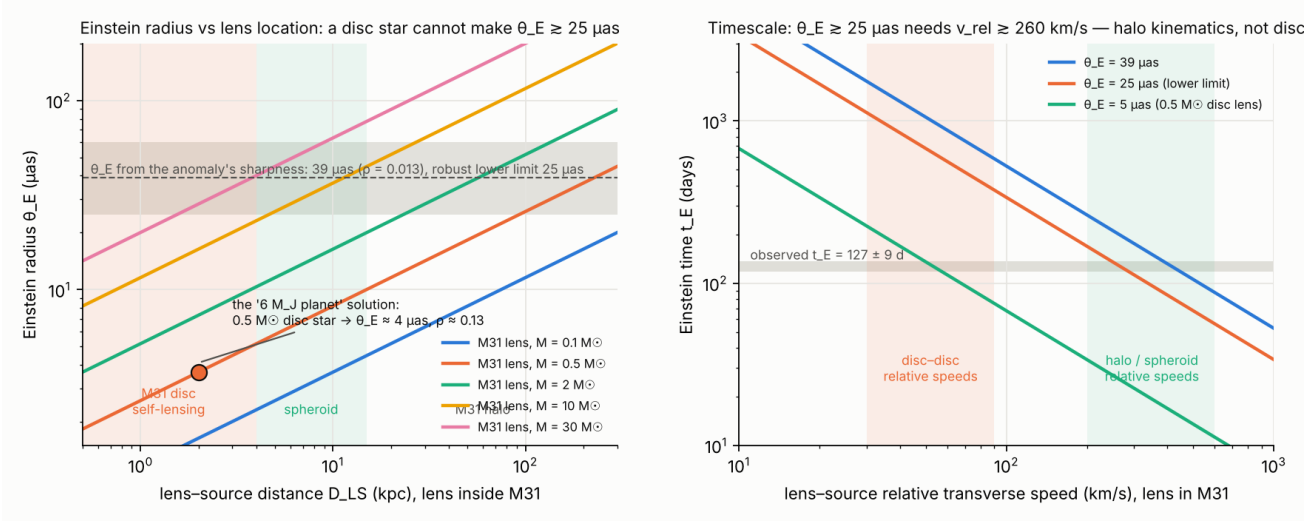


Figure 5: Left: Einstein radius against lens–source distance for lenses inside M31; the band is the $\theta_{\rm E}$ implied by the sharpness of the anomaly. Right: Einstein time against relative transverse speed; disc kinematics cannot produce $t_{\rm E} = 127$ d with $\theta_{\rm E} \geq 25\,\mu$as.

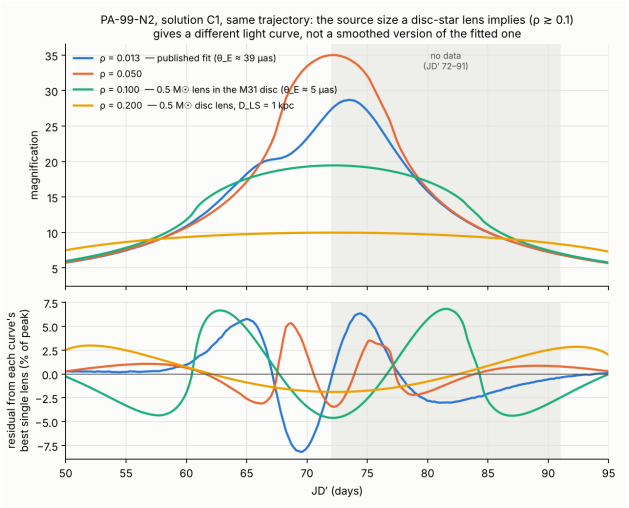


Figure 6: Finite-source light curves for the PA-99-N2 solution C1 at the published ρ and at the values implied by a $0.5\,M_\odot$ disc lens, and residuals from each curve's best single-lens fit.

Table 1: PA-99-N2: probabilities under three treatments of the finite-source constraint. Ranges span halo MACHO fractions 0.01–0.2 with an agnostic (log-uniform) MACHO mass prior.

ρ treatment	P(lens in M31)	$P(M_{\rm c} < 13\,M_{\rm J})$	P(planet in M31)
ignored	0.96	0.66–0.77	0.62–0.73
upper limit only	0.31–0.70	0.31–0.74	0.002–0.012
as published	0.79–0.95	0.12–0.31	0.02–0.05

probability of a planet in M31 is 2–5%. With only the upper limit, the most probable *planetary* solution is a foreground Galactic M dwarf at ~ 2 kpc with a $\sim 4\,M_{\rm J}$ companion—a planet, but a Milky Way one. The single scenario that restores a planet in M31 assumes a population of $0.5\,M_\odot$ MACHOs; the data then favour a 4–7 $M_{\rm J}$ companion to a dark object 100–200 kpc from either galaxy, with comparable odds for the two haloes, at a MACHO fraction excluded in the Milky Way by EROS and OGLE (Tisserand et al., 2007; Mróz et al., 2024).

4.4 Verdict

The binary-lens anomaly is probably real (it improves on a single lens by $\Delta\chi^2 = 512$ and is seen in independent MEGA photometry), though $\chi^2/{\rm dof} = 0.66$, a 4%-per-arcsecond seeing systematic and parallax alternatives within $\Delta\chi^2 \simeq 25$–30 counsel caution. Its mass ratio is well determined. Its planetary reading requires a lens that the same light curve excludes. The lens is most probably a stellar-mass black hole in the M31 halo or spheroid with a low-mass stellar companion.

5 Quasar-microlensing "planet populations"

5.1 The claim

Dai & Guerras (2018) examined 38 Chandra observations of the quadruply lensed quasar RX J1131−1231 ($z_{\rm s} = 0.658$, $z_{\rm l} = 0.295$), in which Chartas et al. (2017) had fitted narrow Gaussian lines to each image's spectrum and found Fe Kα energies shifted from 6.4 keV, with $g = E/6.4$ spanning 0.5–1.3. Shifts significant at 3σ occur in 26%, 34%, 21% and 11% of epochs for images A–D. Assuming that each shifted line marks a caustic crossing the Fe Kα region ($\lesssim$

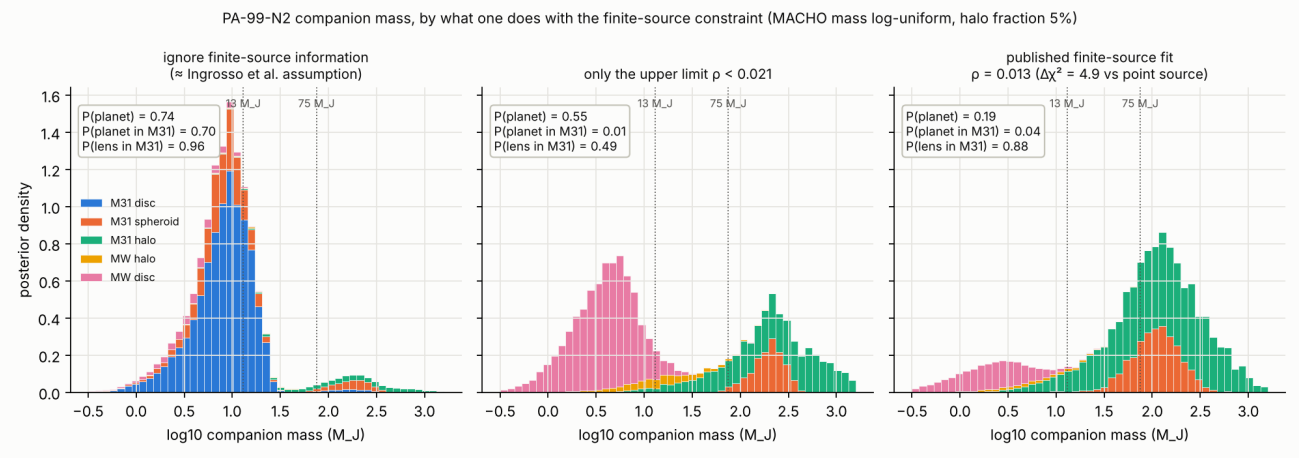


Figure 7: Companion-mass posterior for PA-99-N2, stacked by lens population, for the three treatments of the finite-source constraint (MACHO mass log-uniform, halo fraction 5%).

$10\,r_g \simeq 1.9 \times 10^{14}$ cm), they computed the probability that a caustic crosses a source of that size at a random time from magnification maps with stars only ($\kappa_*/\kappa \simeq 0.1$) and with an added population of free-floating bodies, and found that stars alone are excluded at 4.5–6.4σ while a planet surface-mass fraction $\alpha_{\rm pl} \gtrsim 10^{-4}$ of the total is required, which they expressed as $\gtrsim$ 2000 objects from Moon to Jupiter mass per main-sequence star. Bhatiani et al. (2019) applied the same method to Q J0158−4325 (10 lines in 12 spectra) and SDSS J1004+4112 (8 in 40).

5.2 The null model

The logic treats every shifted narrow line as an event. But RX J1131−1231 possesses a strong *broad* relativistic Fe Kα line in its stacked spectrum; it is the object in which Reis et al. (2014) measured a black-hole spin from that line, and its g range for a disc seen at $i \simeq 70$–$76°$ is precisely 0.5–1.3. The individual spectra have 150–7400 counts (B–D) and up to a few thousand for most epochs of A. A narrow Gaussian of free energy fitted to such a spectrum will lock on to noise peaks within the broad hump, and its energy uncertainty from a $\Delta C = 1$ interval will be a few tens of eV.

We simulated static spectra (power law $\Gamma = 1.7$; a narrow 6.4 keV core of rest-frame equivalent width 150 eV, the value Chartas et al. (2017) find in the stacked image-C spectrum; a broad line from a Keplerian disc at $i = 70°$ with emissivity r^{-3} or r^{-6} and rest EW of 300–600 eV), folded through an ACIS-like response, and fitted each as in Chartas et al. (2017): C-statistic, power-law null, power law plus unresolved Gaussian of free energy over rest 3.5–9 keV. The 90% and 99% detection thresholds were calibrated on line-free simulations, which includes the look-elsewhere effect. A detected line was called "shifted" if $|g - 1|$ exceeded 3 (or 1) times its $\Delta C = 1$ uncertainty.

The results (Fig. 8) at 1000–3000 counts per spectrum are: lines detected at $> 90\%$ in 28–64% of spectra (observed 51%), at $> 99\%$ in 6–27% (observed 14%), 3σ-shifted narrow lines in 15% (core plus broad) to 19–31% (broad only), against the observed 21–34%, with shifted energies spanning $g = 0.55$–1.3 (observed 0.5–1.3) and the large fitted equivalent widths that Chartas et al. (2017) report. An independent implementation with a Gaussian broad line found 13–17% (core plus broad) and 17–30% (broad only) at 1000 and 3000 counts, and its line-free control gave 3σ-"shifted" lines in 8–9% of spectra, as it must: a 90% threshold admits 10% false lines, and essentially all of them are shifted. No single static model reproduces all three observed fractions simultaneously; the actual spread of counts across 152 spectra, and the stellar microlensing that is undisputedly present ($\kappa_*/\kappa \simeq 0.1$; this quasar's X-ray flux ratios vary by factors of several), bracket the observations without difficulty.

5.3 Verdict

The statistic from which the planet population was inferred has a null expectation of 15–30%, not the few per cent implied by counting stellar caustic crossings alone. Nothing remains for planets to explain. Two further points stand independently of the simulation. First, the quoted mass range is not what the data probe: a Moon-mass lens has a source-plane Einstein radius of 9×10^{12} cm against a $\gtrsim 2 \times 10^{14}$ cm emitting region and perturbs its magnification by $\sim 10^{-3}$; only bodies above a few Earth masses can shift the line, and the "2000 per star" number is dominated by the extrapolated low-mass end of an assumed $N \propto M^{-2}$ distribution. Second, the implied free-floating mass per star ($\gtrsim 1\,M_{\rm J}$) exceeds the Galactic microlensing measurements by a factor of several (Mróz et al., 2017; Sumi et al., 2023). The 2019 systems inherit all of this with far fewer spectra.

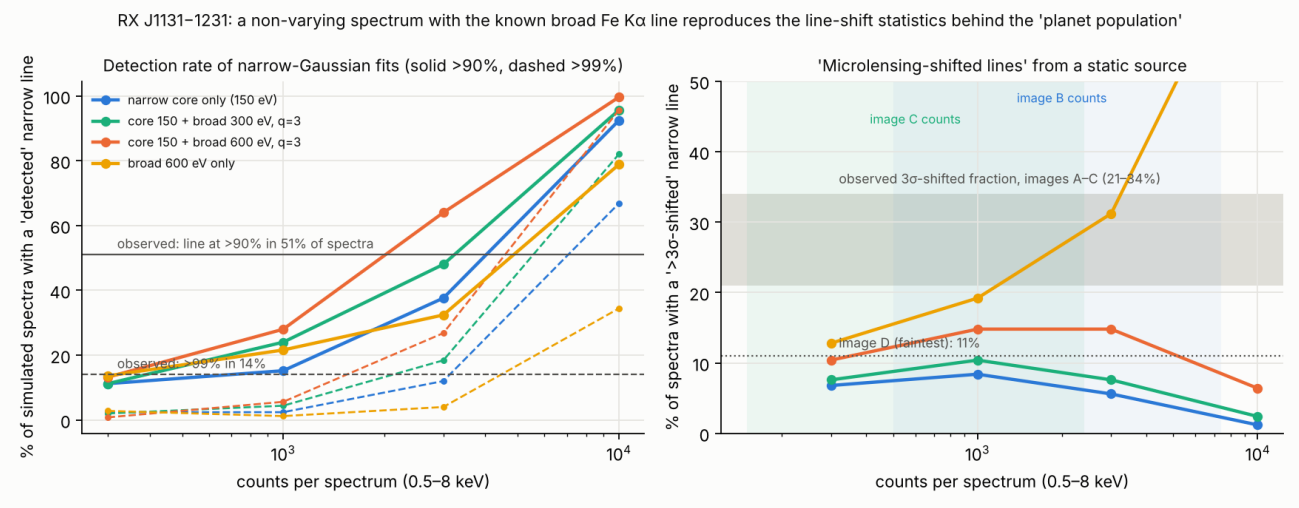


Figure 8: Left: fraction of simulated static spectra in which a narrow-Gaussian fit detects a line at $> 90\%$ (solid) and $> 99\%$ (dashed) confidence, against counts per spectrum, for four line models. Right: fraction with a $> 3\sigma$-shifted line. The observed fractions for RX J1131−1231 are marked.

6 Single-event quasar microlensing

6.1 Q0957+561

Schild (1996) interpreted rapid (∼day) fluctuations of $\sim 1\%$ in the time-delay-corrected difference of the two Twin Quasar images as microlensing by planetary-mass bodies in the lens galaxy, popularly reported as a 3 Earth-mass planet; Colley & Schild (2003) later reported a 0.93%, 12-hour event. A point lens whose Einstein radius is much smaller than the source changes the source's magnification by at most $\simeq 2(R_{\rm E}/R_{\rm s})^2$. For Q0957+561 ($z_{\rm l} = 0.355$, $z_{\rm s} = 1.41$) the source-plane Einstein radius of $3\,M_\oplus$ is 1.8×10^{14} cm; the R-band disc is 5×10^{15}–10^{16} cm (Refsdal et al., 2000; Hainline et al., 2012). The maximum perturbation is 0.07–0.26% and the event lasts $2R_{\rm s}/v \simeq 3$–6 yr at $v = 1000\,{\rm km\,s^{-1}}$ (Fig. 9). A 1% event in half a day requires $\sim 10^{-7}\,M_\odot$ lenses crossing $\sim 10^{13}$ cm brightness structures in the disc, which is the ad hoc "double-ring" quasar of Colley & Schild (2003). The 2003 event itself rests on 12 overlapping points from two epochs 14 months apart, and the authors' leave-one-night-out significances are 41–74%.

6.2 Q2237+0305

The 1989 brightening of image A of the Einstein Cross (Irwin et al., 1989) was the first detection of quasar microlensing and is universally interpreted as a stellar caustic crossing in the lens bulge ($\kappa_* \simeq 0.4$–0.8). The "1.6 Jupiter-mass" reading that has entered reference lists appears in no paper: it is $R_{\rm E} = v\,t$ with $t = 420$ d and $v \simeq 2000\,{\rm km\,s^{-1}}$. The effective source-plane velocity for this system is $(D_{\rm s}/D_{\rm l}) \simeq 11$ times the lens-plane velocity, i.e. 3000–6500 km s^{-1} (Kochanek, 2004; Mosquera & Kochanek, 2011), which in the same reading gives 10–30 $M_{\rm J}$; and the reading is wrong in kind, because the duration of a caustic crossing measures the source size ($R_{\rm s} \simeq vt \simeq 1$–$5 \times 10^{15}$ cm, which is the measured disc size), not the lens mass. Tuntsov (2024) tests a population of $10^{-4}\,M_\odot$ "nanolenses" making up 6–11% of the lens convergence against time-derivative statistics of filtered OGLE light curves, describes the outcome as "not much more than suggestive", gives no significance, and notes that a 10^{16} cm source removes the signal. Such a population, if compact, is excluded by Galactic constraints on free-floating bodies by two orders of magnitude.

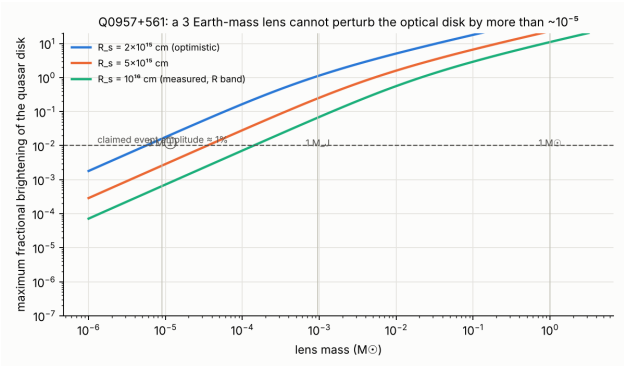


Figure 9: Maximum fractional brightening of the Q0957+561 optical disc by a single point lens, against lens mass, for three source radii.

7 FRB 20180916B

Kurban et al. (2022) proposed that the 16.35-day periodicity of FRB 20180916B arises from a $10^{-5}\,M_\odot$ ($3.3\,M_\oplus$) rocky planet on an $e = 0.88$–0.92 orbit about a neutron star, partially disrupted at each periastron ($r_{\rm p} = 2r_{\rm td} \simeq 1.6$–$2.4 \times 10^{11}$ cm); km-sized debris clumps then interact with the pulsar wind through Alfvén wings (Mottez & Zarka, 2014), and their orbital-

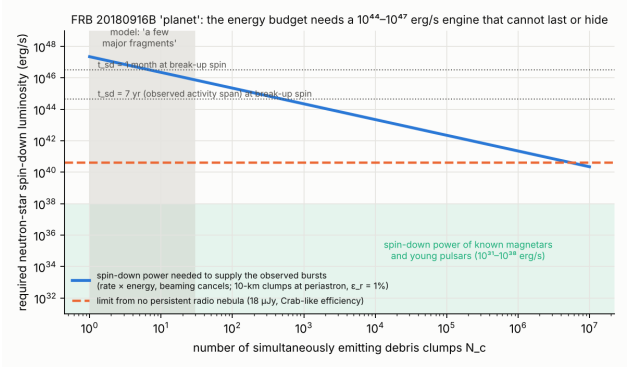


Figure 10: Spin-down luminosity required by the FRB 20180916B planet model, against the number of emitting clumps, compared with spin-down lifetimes, the persistent-source limit and known pulsars.

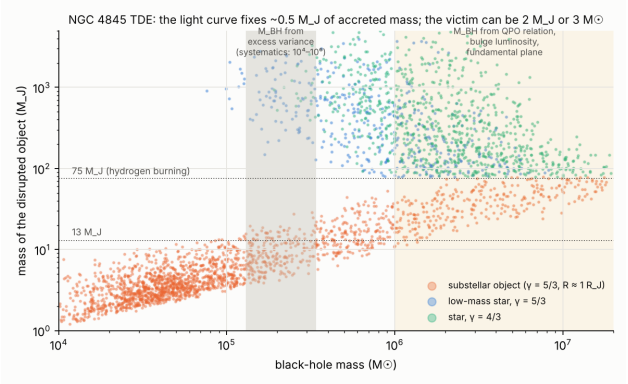


Figure 11: Victim masses consistent with the NGC 4845 flare, against black-hole mass, for substellar and stellar victims; shaded bands show the black-hole-mass ranges from the different estimators.

period spread sets a $\sim$ 5-day active window. The model's flux formula multiplies a $4\pi/\Omega$ beaming factor ($\Omega = 0.1$ sr) by an additional $(\gamma/10^5)^2$ with $\gamma = 3\times10^6$.

Beaming cannot help a time-averaged budget. The observed burst rate times the observed isotropic energy equals the true power emitted by all clumps, because each clump's beaming factor cancels between the two. CHIME measures a peak rate of $0.86^{+0.5}_{-0.35}$ h^{-1} above 5.2 Jy ms with a fluence index of -2.3 (Sand et al., 2023); with a 400 MHz band at 150 Mpc the mean burst energy is 2.4×10^{38} erg and $\langle L_{\rm iso}\rangle = 5.8\times10^{34}\,{\rm erg\,s^{-1}}$ at phase peak. With the model's radio efficiency $\epsilon_r \le 10^{-2}$ the clumps must tap $\ge 5.8\times10^{36}\,{\rm erg\,s^{-1}}$. The Alfvén-wing power of a clump of radius R_c at distance r is $1.5(R_c/r)^2 = 2.6\times10^{-11}$ of the spin-down luminosity for 10 km clumps at periastron, so

$$L_{\rm sd} \;\ge\; \frac{2.2\times10^{47}}{N_c}\ {\rm erg\,s^{-1}}, \qquad (2)$$

where N_c is the number of simultaneously emitting clumps; the model produces "a few major fragments". A neutron star at break-up spin holds $\lesssim 10^{53}$ erg, so this engine lasts 5 days ($N_c = 1$) to 14 yr ($N_c = 10^3$), against a period that has remained 16.34 ± 0.07 d with a constant Poisson rate since 2018 (Sand et al., 2023); and any such wind would feed a nebula 5×10^3–5×10^6 times brighter than the 18 μJy persistent-source limit of Marcote et al. (2020) (Fig. 10). Tidal heating at $r_{\rm p} = 2r_{\rm td}$ also deposits the planet's binding energy in $\sim$ 200 yr. The chromatic activity window—earlier at high frequency, later at 150 MHz by $\sim$ 3 d (Pastor-Marazuela et al., 2021; Pleunis et al., 2021)—has no counterpart in a debris-clump model. The independent re-derivation reproduced every number in this section.

8 The NGC 4845 tidal disruption event

IGR J12580+0134 was a hard X-ray flare in the nucleus of NGC 4845 (January 2011) with a $t^{-5/3}$ decline, peak 17–80 keV luminosity $1.5\times10^{42}\,{\rm erg\,s^{-1}}$ and $t_{\rm peak} = 60$–100 d. Nikołajuk & Walter (2013) derived a black-hole mass of $2.3^{+1.1}_{-1.0}\times10^5\,M_\odot$ from excess variance (systematic range 10^4–10^6) and, with the fallback fits of Guillochon & Ramirez-Ruiz (2013), a 10% radiative efficiency, a bolometric correction of 10 and a partial disruption releasing $\sim$ 10% of the victim, a "14–30 $M_{\rm J}$ super-Jupiter"; Lei et al. (2016) quote 8–40 $M_{\rm J}$ over 3×10^5–$1.2\times10^7\,M_\odot$, while the other mass estimators they list (QPO relation, bulge luminosity, fundamental plane) give 10^6–$2\times10^7\,M_\odot$.

The two observables fix only the product $\dot M_{\rm peak}t_{\rm peak}$, i.e. the accreted mass, $\simeq 0.5\,M_{\rm J}\,(0.1/\eta)(f_{\rm bol}/10)$. We inverted $t_{\rm peak}$ and $\dot M_{\rm peak}$ with the Guillochon & Ramirez-Ruiz (2013) scalings over $\eta \in [0.06, 0.4]$, $f_{\rm bol} \in [1, 10]$, $M_{\rm BH}$ log-uniform in $[10^4, 2\times10^7]\,M_\odot$, impact parameters β across the simulated range, an Eddington ceiling of $3L_{\rm Edd}$, and mass–radius relations for substellar objects and stars. Acceptable victims are substellar objects of 2–46 $M_{\rm J}$ (62% of solutions at equal class prior) or stars of 0.1–3 $M_\odot$ stripped of < 1% of their mass at $\beta \simeq 0.5$–0.7 (38%); the stripped mass is 0.4–0.8 $M_{\rm J}$ in every case (Fig. 11). Forcing the excess-variance black-hole mass narrows the substellar branch to 4.5–19 $M_{\rm J}$ but leaves the stellar one at $\sim$ 25%, and a galactic nucleus offers far more stars than super-Jupiters on loss-cone orbits. The 2023 NICER detection of weak flares from the same nucleus (arXiv:2504.18558, 2025) shows it to be a low-luminosity AGN, so even the 2011 event's classification rests on the decline law alone. The super-Jupiter is one assumption-dependent corner of a degeneracy that spans three orders of magnitude.

9 Milky Way stars of extragalactic origin

BD+20 2457. Niedzielski et al. (2009) reported two companions with $m \sin i = 21.4$ and $12.5\,M_{\rm J}$ at 1.45 and 2.01 AU ($e = 0.15$, 0.18) around a $2.8\,M_\odot$ giant; Perottoni et al. (2021) suggested the star formed in Gaia–Enceladus. The pair is separated by 2.1 mutual Hill radii, below the $2\sqrt{3}$ required for Hill stability. REBOUND integrations (Rein & Liu, 2012) of the published system end in ejection or $e > 0.98$ within 35–2500 yr over 16 orbital-phase combinations, 275 yr for circular orbits, 828 yr with halved masses and 20 yr for a $1.5\,M_\odot$ star, confirming Horner et al. (2014); the independent run found 4–270 yr. Whatever produces the radial velocities of this gigayear-old giant, it is not that pair of planets.

HIP 13044 b. The $1.25\,M_{\rm J}$ planet around this Helmi-stream star (Setiawan et al., 2010) was withdrawn after Jones & Jenkins (2014) showed the signal arose from an erroneous barycentric correction; the corrected data show no planet.

HD 134440. Reggiani & Meléndez (2018) inferred past planet engulfment from an abundance difference between HD 134440 and HD 134439. This is a chemical inference, not a planet detection, and falls outside the scope of refutation.

10 Discussion

Four failure modes recur.

The favourable reading of an unconstrained scale. A single event determines a dimensionless combination—an occulter-to-source size ratio, a lens mass ratio, an accreted mass, a timescale—and the physical quantity of interest follows only after a scale is supplied: the lens mass (PA-99-N2), the black-hole mass (NGC 4845), the effective velocity (Q2237+0305), the number and size of debris clumps (FRB 20180916B). In each case the published planet is the value obtained for one choice of scale, and other choices within the published uncertainties give a star, a black hole or nothing.

The absent null. The quasar-microlensing populations were inferred from an excess over a null that counted only one mechanism (stellar caustics) and omitted the dominant ones (a static broad line sampled by narrow fits in low-count spectra, and the false-detection floor of the threshold itself). The M51-ULS-1b hardness test and shape test were asserted rather than computed; computed, they carry odds of a few to one.

Detection without conditioning. An event found in a search of known sensitivity must be weighted by the probability of being found. For M51-ULS-1b this moves the posterior from a Saturn at 45 AU to an M dwarf at 2 AU and exposes a rate excess of 10^2–10^3; for PA-99-N2 it is what makes halo remnants, not disc stars, the most probable lenses.

Internal inconsistency. PA-99-N2's assumed lens predicts a source size the same light curve excludes; the FRB planet's assumed engine would have spun down before the second observing season.

A confirmed extragalactic planet will require a measurement that breaks the scale degeneracy within the event itself: a microlensing event towards M31 with both finite-source and parallax signatures, giving $\theta_{\rm E}$ and $\pi_{\rm E}$ and hence the lens mass directly; or an X-ray transit with enough photons in ingress and egress to distinguish a central crossing from a grazing one ($\Delta \ln L \gtrsim 5$, roughly seven times the M51-ULS-1b count rate), together with a measured base rate of look-alike dips in the same source class. Rubin/LSST pixel lensing of M31 and the Roman microlensing survey of the Bulge, with their dense sampling, are the plausible routes; archival Chandra data cannot provide the first and have not provided the second.

11 Summary

Table 2 collects the outcomes. No entry on the list of extragalactic planets has been observed: each is a single unrepeatable event read through a favourable assumption, a statistic whose null model was not computed, or a claim already withdrawn. None of this bears on whether planets exist in other galaxies; it bears on whether any has yet been detected.

Acknowledgements

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Data availability

All code (light-curve reconstruction and fits for M51-ULS-1b; binary-lens ray shooting, finite-source refits and lens-population model for PA-99-N2; the Fe Kα spectral simulation; the FRB energy budget; the tidal-disruption inversion; the N-body integrations), the numerical results in machine-readable form, and the independent verification runs are provided as supplementary material and will be deposited in a public repository on acceptance. Published parameters were taken from the cited papers; no proprietary data were used.

Table 2: Status of all published extragalactic-planet candidates after this work.

Candidate	Claim	Status	Decisive point
M51-ULS-1b	$0.7\,R_{\rm J}$ transiting planet	unconfirmable; planet disfavoured	dip does not measure size; rate excess 10^2–10^3
PA-99-N2	$6.3\,M_{\rm J}$ companion	refuted as published	needs $\rho \simeq 0.1$; light curve requires $\rho < 0.02$
RX J1131−1231	$\gtrsim 2000$ planets per star	unsupported	shift statistic reproduced by a static spectrum
Q J0158, SDSS J1004	same	unsupported	same method, 12 and 40 spectra
Q0957+561	$3\,M_{\oplus}$ body	refuted	$\leq 0.3\%$ over years, not 1% in a day
Q2237+0305	$1.6\,M_{\rm J}$; nanolens population	dismissed	misread caustic crossing; "suggestive"
FRB 20180916B	$2.9\,M_{\oplus}$ planet	refuted	energy budget short by $10^{3.7}$–10^{7}
NGC 4845 TDE	14–$40\,M_{\rm J}$ victim	unsupported	victim mass degenerate, $2\,M_{\rm J}$–$3\,M_{\odot}$
BD+20 2457 b, c	two super-Jupiters	refuted as published	system disintegrates in 4–2500 yr
HIP 13044 b	$1.25\,M_{\rm J}$	refuted (Jones & Jenkins, 2014)	barycentric-correction error
HD 134440	engulfment	not a detection	—

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